Unit 3 Test Version B Teacher Companion

Teacher use · answer key, scoring and instructional response · 19 points

Answer key and partial credit

Item	Expected answer	Suggested scoring
1	20 blue beads; 4 red beads.	1: both quantities correct.
2A	30/36 = 5/6.	1: equivalent ratio in simplest form.
2B	For every 5 students in robotics, there are 6 on the soccer team.	1: correct order and quantities.
3	4 jumping jacks per second.	1: numerical rate and correct unit.
4A	$5.65/lb at Fresh Market; $5.40/lb at Nature's Pantry.	1: both correct; 0.5 each.
4B	Nature's Pantry, because its unit price is lower (second option).	1: correct choice and reason.

Answer key and partial credit

Item	Expected answer	Suggested scoring
5	Tasha earns $3 more per hour: 56 ÷ 4 = 14; Devon earns $11/hour.	1: Tasha and 3; 0.5 each.
6A	White milk: 9, 15, 24, 33.	1: 0.25 per entry.
6B	Multiply chocolate counts by 3/4, or divide by 4 and multiply by 3.	1: explanation maintains the 4:3 relationship.
7	(1,3), (2,6), (3,9), (4,12), (5,15).	1: 0.2 per correctly plotted point.
8A	A: (1,3),(2,6),(3,9),(4,12). B: (1,4), (2,8),(3,12),(4,16).	1: 0.125 per correctly completed pair.
8B	Brand B: 4 g/serving versus 3 g/serving.	1: Brand B.

Answer key and partial credit

Item	Expected answer	Suggested scoring
9	9 minutes per mile; 5 miles in 45 minutes.	1: 0.5 per blank.
10	Equivalent: 270/30 = 180/20 = 9 calories/minute.	1: 0.5 for each justified equal unit rate; an unsupported yes earns 0.
11	132 pounds (fourth option), using 1 kg ≈ 2.2 lb.	1: correct choice.
12A	Each ratio part is 42 ÷ 7 = 6. Bagels = 4 × 6 = 24. Subtracting 3 treats the ratio as a fixed difference; at this scale the difference is 18.	2: 1 for identifying additive reasoning; 1 for the multiplicative solution with 24 bagels.
12B	Label every equal box 6; muffins total 42 and bagels total 24.	1: consistent box values and correct totals.
13	700 yards.	1: 2100 ÷ 3 = 700.

Scoring decisions

These suggested partial-credit rules total 19 points. Apply them consistently. Accept mathematically equivalent explanations and documented response accommodations. Do not double-penalize a correct interpretation in 2B solely because a simplification error occurred in 2A. Review arithmetic slips separately from ratio or unit reasoning.

Use results for reteaching

1–2: ratio meaning and order. 3–5: unit rates and comparisons. 6–8: equivalent ratios across tables and graphs. 9–10: rate reasoning. 11 and 13: unit conversion. 12: multiplicative versus additive reasoning. Select the corresponding lesson Teaching Support Guide and reassess with an unused alternate-form item.